

refracted light
A simple triangular glass prism will split white light (or sunlight) into the colors of the rainbow. We now know that each color corresponds to a different band of wavelengths, and the shorter the wavelength, the greater the refraction.



refraction of white light splits it into its constituent wavelengths.....

The icing on the cake came when Maxwell used his equations to calculate the speed at which an electromagnetic wave must travel. The equations give a unique speed for the wave, and for all electromagnetic waves. This speed turned out to be exactly the speed of light. All electromagnetic waves travel at the speed of light, and the inescapable conclusion is that light consists of an electromagnetic wave traveling through space.

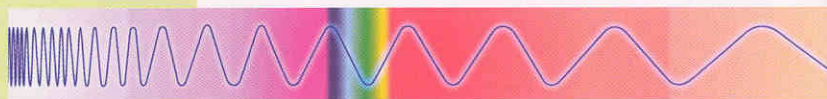
light wave problems

It was already known that light travels through space in the form of a wave from the work of Thomas Young in England and Augustin Fresnel in France early in the 1800s. Faraday had suggested that light is "a high species of vibration in the lines of force," but he lacked the mathematical knowledge to describe how this could work.

Experiments like the ones carried out by Young show that visible light has a wavelength (the distance between two successive peaks of the waves) in the range from about 380 nanometers (violet) to 750 nanometers (red). The same equations were soon being used to describe longer wavelength radiation (such as radio waves) and shorter wavelength radiation (ultraviolet light and X-rays). However, there was one problem with the wave theory of light. It could not explain a phenomenon known as black body radiation. At first, this seemed to be a minor irritation that could soon be resolved, but the more physicists looked at the problem, the bigger it grew.

electromagnetic spectrum

Visible light makes up just a small portion of the electromagnetic spectrum, which ranges from very short wavelength gamma radiation to radio waves. The extent to which an electromagnetic wave is refracted depends on its wavelength.



gamma x-ray ultraviolet visible light infrared microwave radio wave

the quantum revolution

Black body radiation gets its name in a curiously backward way. In classical physics, a black body is an object that absorbs all the electromagnetic radiation that falls on it. If such an object were to become hot, it would

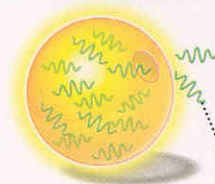
"The only laws of matter are those which our minds must fabricate, and the only laws of mind are fabricated or it by matter."

James Clerk Maxwell

then radiate energy—black body radiation—but it would no longer be black. Its old name—cavity radiation—gives a much more telling insight into the nature of this kind of radiation.

what is black body radiation?

Imagine a large, hollow sphere in which there is one tiny hole. Any radiation that arrives at the hole goes through and is absorbed inside the cavity, which acts like a black body. Now imagine heating the sphere up until it glows, first red-hot, then white-hot, then blue-hot. The radiation that comes out of the hole is pure black body (or cavity) radiation. The name "black body" is applied, somewhat confusingly, even to this colored radiation. This example highlights one of the most important features of black body radiation. Its color depends on its temperature. Since the color of light is related to its wavelength, this means that the intensity of the radiation



key points

- Light travels as a wave, like ripples on a pond.
- Newton's laws and Maxwell's equations seemed to describe almost everything about the physical world at the end of the 1800s.
- Those laws could not explain the nature of light radiated by hot objects.

absorbing and radiating
A cold sphere with a hole in it absorbs electromagnetic radiation and acts like a black body. When it gets hot, it emits electromagnetic radiation. This is black body radiation.

radiation escapes from cavity